

Huffman Coding

Message $\rightarrow$ BCCABBDDEAECCBBAEDDCC

length = 20

ASCII - 8-bit

$$8 \times 20 = 160 \text{ bits}$$

A 65 01000001

B 66 01000010

C 67 ' .

D 68 ' .

E 69 ' .

Huffman Coding

Message $\rightarrow$ BCCABBDDEAECCBBBAEDDCC
001010 . . .

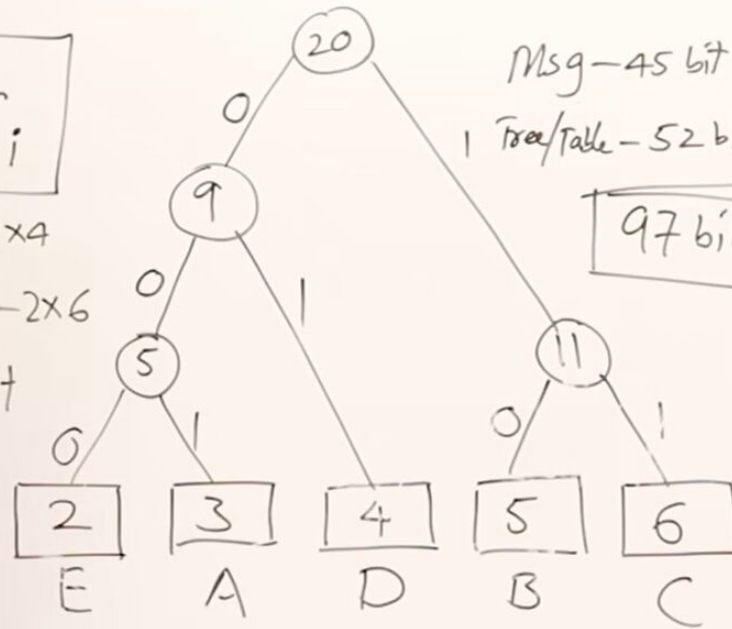
character	count/frequency	Code	$20 \times 3 = 60 \text{ bits}$
A	3 $3/20$	000	$\frac{5 \times 8 \text{ bit}}{\uparrow \text{ character}}$ $\frac{5 \times 3}{\uparrow \text{ codes}}$
B	5 $5/20$	001	
C	6 $6/20$	010	$40 + 15 = 55$
D	4 $4/20$	011	Msg — 60 bits
E	2 $2/20$	100	Table — 55 bits
	20		<u>115 bits</u>

Huffman Coding

Message $\rightarrow$ BCCABBBDDAECCBBBAEDDCC
 10 11 11 00 10 10 01 01 - - -

$$\sum d_i * f_i$$

$$3 \times 2 + 3 \times 3 + 2 \times 4 + 2 \times 5 + 2 \times 6 = 45 \text{ bit}$$



Msg - 45 bit

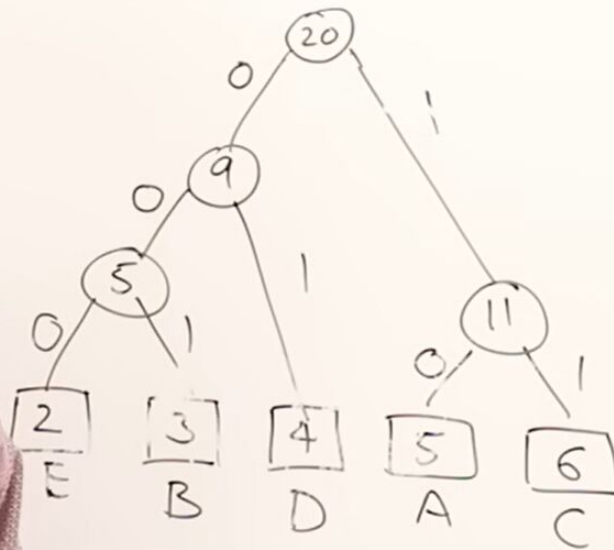
Tree/Table - 52 bits

97 bits

Char	Count	Code	
A	3	001	$3 \times 3 = 9$
B	5	10	$5 \times 2 = 10$
C	6	11	$6 \times 2 = 12$
D	4	01	$4 \times 2 = 8$
E	2	000	$2 \times 3 = 6$
5 x 8 bit		20	45 bits
40 bits		+	
		12 bit	

Huffman Coding

Message → BCCDACCBDABCCDEAAEDA
001011100101100111101 - - - -



BCCD